

## Education

### Texas A&M University

*Pursuing M.S. Computer Science*

College Station, TX

*Expected 2028*

### Texas A&M University

*B.S. Computer Science & B.S. Statistics, **Summa Cum Laude***

College Station, TX

*May 2026 | GPA: 3.92/4.00*

## Technical Skills

**Data Science & Statistics:** Python, R, SQL, pandas, NumPy, SciPy, scikit-learn, Bayesian inference, MCMC, hypothesis testing, FDR correction, Matplotlib

**Bioinformatics & HPC:** GenomeMap, PAML/codeml, dN/dS, pN/pS, FASTA pipelines, Slurm, Linux, TAMU HPRC

**Machine Learning & Data Systems:** PyTorch, Hugging Face Transformers, LangChain, AWS, MongoDB, Elasticsearch, Redis, Git, GitHub Actions

**Programming:** Java, C++, TypeScript, JavaScript, React, FastAPI, Docker, PostgreSQL

## Experience

### Graduate Assistant Research (GAR) - Computational Genomics

June 2026 - Present

*Ioerger Lab, Texas A&M University*

*College Station, TX*

- Bayesian dN/dS with **GenomeMap** on *M. tuberculosis*, including lineage comparisons (L1–L4) and a diabetes comparison of **922** isolates (**178** and **744**).
- Built DPD from the MCMC samples: the probability that selection in one group is higher than in the other. Parallelized the jobs as **Slurm** array tasks.
- Fit gene-level linear models to rifampicin TnSeq fitness and tested them with likelihood-ratio tests. Checked the same genes with **PAML/codeml** and pN/pS.

### Software Engineering Capstone - Amazon-Sponsored

August 2025 - December 2025

*Texas A&M University*

*College Station, TX*

- Built a breach-intelligence analytics platform in a **3-person team**, implementing **Scrapy** and **Tor** ingestion plus a **LangChain** and **Gemini 2.5 Flash Lite** pipeline for entity extraction and threat classification.
- Designed storage and retrieval across **AWS S3**, **MongoDB**, **Elasticsearch**, **Redis**, and **Neptune**; delivered Streamlit and Kibana views for querying unstructured breach data by sector and geography.

### Undergraduate Researcher

January 2025 - May 2025

*UrbanResilience.AI Lab, Texas A&M University*

*College Station, TX*

- Built **Python** ingestion and preprocessing pipelines integrating air-quality sensor APIs and web-scraped wildfire data across Los Angeles sources for downstream emergency-response analysis.

### AI Engineering Intern

May 2024 - August 2024

*Videomagic*

*Remote*

- Developed and evaluated **PyTorch** and **Hugging Face Transformers** computer-vision workflows for AI-generated video detection, analyzing frame-level outputs and model failure modes for product deployment.
- Built **Feathers.js**, **MySQL**, and **Knex.js** APIs for model inference and data management, secured with **Auth0** and **JWT RBAC**; shipped AI features associated with **15% higher engagement**.

## Projects & Competitions

### PickLedgerPro | *Python, scikit-learn, pandas/NumPy, TypeScript, GitHub Actions*

- Shipped a public pick board from committed JSON: **8 team-model** and **3 player-prop** pipelines, an immutable pregame ledger, and ESPN grading every **15 minutes**.
- Score settled picks with **Brier**, **log loss**, and **ECE** against the price recorded before the game.

### Alpha - Tidal Hackathon 2025 | *React, TypeScript, Multi-LLM - 1st Place*

- Built a math-learning SPA orchestrating Gemini, OpenRouter, Claude, Manim, Desmos, and Wolfram Alpha; reduced tool-call failures from **25% to 6%** through output normalization and validation.